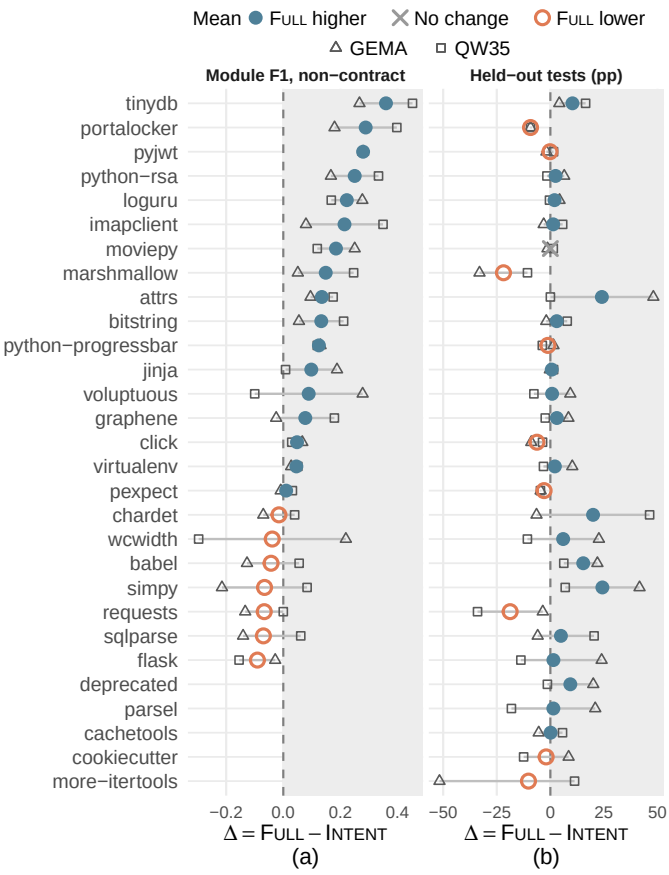

Fig. 3. Per-library gain of FULL over INTENT in (a) module F1 on the modules the contract does not name and (b) held-out tests passed (pp). Large dots: mean over generators; small markers: each generator. Shaded side: FULL higher. Sorted by mean module F1 gain; the 5 libraries at the bottom have no module F1, as the contract names every module.

generators (Table VI): no significant change in correctness or co-location agreement, and a significantly higher module F1. Neither the generator (Gen., i.e., whether outcomes differ between generators in both conditions alike) nor its interaction with FULL (Int., i.e., whether the effect of FULL over INTENT differs between generators) is significant for any primary outcome. The gain in module F1 is larger with QW35 (Figure 3), whose artefacts name more of the library's modules, but the difference is not significant (Int. $p = 0.484$), and on many libraries the generators disagree on the direction of the effect.

5) Contract coverage: The contracts name between 7% and 100% of a library's modules (median 67%). If the contract masked the benefit of the artefacts, the gain in module F1 would shrink as contract coverage grows. There is no sign of this: the correlation is positive, though not significant ($\rho = +0.29$, $p = 0.162$). Splitting the libraries at the median gives the same picture: the gain is +0.05 below and +0.15 above, and the condition × coverage interaction is not significant ($p = 0.099$). For held-out tests the gain is larger below the median (+3.1 vs +1.2 pp), but neither the correlation ($p = 0.214$) nor the interaction ($p = 0.230$) supports a dependence.

TABLE VII
GAIN OF FULL (p IN PARENTHESES). CONTRACT COV.: SHARE OF MODULES THE CONTRACT NAMES; ARTEFACT COV.: SHARE OF NON-CONTRACT MODULES THE ARTEFACTS NAME. β: WITHIN LIBRARY.

		Module F1 (24)	**Held-out (pp)** (29)
Contract cov.	Spearman ρ	+0.29 (.162)	−0.24 (.214)
	Gain < / > median	+0.05 / +0.15	+3.1 / +1.2
	Interaction p	.099	.230
Artefact cov.	Spearman ρ	+0.07 (.756)	−0.24 (.263)
	β per SD	+0.011 (.664)	−0.5 (.843)
	β per SD, length-adj.	+0.004 (.884)	+0.4 (.882)
Length	β per SD	+0.021 (.315)	−3.5 (.125)

6) Artefact coverage and length: Across libraries, the gain does not grow with the share of non-contract modules that the artefacts name ($\rho = +0.07$, $p = 0.756$). Within a library, the generator whose artefacts name more of the modules gains no more module F1: $\beta = +0.011$ per SD of coverage, i.e. per 32 pp ($p = 0.664$), also at equal artefact length ($\beta = +0.004$, $p = 0.884$). Longer artefacts do not help either: $\beta = +0.021$ per SD of length, i.e. per 1.7 k tokens ($p = 0.315$), and −3.5 pp of held-out tests passed ($p = 0.125$).

Within the range of our contracts and generators, what helps the agent is receiving the requirements and architecture, not how many modules they or the contract name.

TAKEAWAY 4 **Concise artefacts suffice.** The gain does not require verbose artefacts: those of the concise generator alone bring the module structure significantly closer to the original (Table VI). The verbose generator names about twice as many modules, yet its gain is not significantly larger, and neither the share of modules named nor the length of the artefacts predicts it. Concise artefacts, also cheaper to generate and to read, carry the structure the agent takes up.

VII. DISCUSSION

Given the requirements and architecture, the agent decomposes the system more like the original. As the gain does not grow with the number of modules the artefacts name, it seems to come from the decomposition they describe rather than from the names alone. In frameworks such as MetaGPT [1] and ChatDev [4], where a designer agent hands its design to a coder, this suggests that the hand-off steers how the coder decomposes the system, and recovered artefacts can supply such a design when building a new system, and even when original ones don't exist.

The agent passes as many tests in both conditions, but this does not mean that design is irrelevant to behaviour. Tests can only run against an interface the agent is given, such as COMMIT0's stubs or our contract, and this interface reveals what will be tested and fixes part of the structure. Whether a design improves correctness when the behaviour is less specified therefore remains open. Few benchmarks target design [5], and test-based ones cannot capture it: in our experiment, the test score alone would have missed the difference. Evaluating design-aware agents requires structural